Low-Power Low-Noise ECG with DRL

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This project presents the design, implementation, and verification of a fully integrated analog front-end (AFE) using the IBM 7RF CMOS process. The goal of this project was to develop a low-noise, high-gain instrumentation architecture that extract microvolt-level biopotential signals in the presence of a large common-mode interference and realistic electric electrode-skin interface conditions. The system is built around a custom differential OTA featuring optimized device sizing, multi-finger transistor layout, and bias current tuning to achieve the required gain-bandwidth tradeoff while minimizing thermal and flicker noise. The Driven-Right-Leg (DRL) circuit was implemented to actively suppress body common-mode potential and improve CMRR. Simulation results confirmed that the designed AFE meets the required gain, bandwidth, common-mode rejection, and noise specifications for ECG signal acquisition.

Introduction

Biopotential amplifiers are critical building blocks in neural and cardiac recording systems. The electrocardiogram (ECG) is a widely used diagnostic tool for assessing cardiac rhythm and conductive abnormalities and is routinely acquired using surface electrodes that sense biopotential differences on the skin [2]. Typical ECH signals are on the order of 0.5-4 mV [1], with most of their energy between a few tenths of a hertz and a few hundred hertz, and are superimposed on large electrode offset voltages (EOVs) and strong power-line interference from the surrounding environment. A practical ECG front end must simultaneously address low-frequency drift, motion artifacts, large 60Hz common-mode interference, and variability in the electrode skin interference.

Front End Architecture

To address these challenges, a single-stage capacitor coupled instrumentation amplifier architecture was utilized. When the low-frequency open-loop voltage gain of the front-end OTA, A_{OL}, is much larger than the capacitive gain factor C_{in}/C_{FB}, the midband closed-loop gain is approximately $A_{v,DM} \approx \frac{C_{in}}{C_{FB}}$, allowing accurate and robust 40dB gain. Large coupling capacitors are used at the input to block electrode DC offsets while preserving the ECG band. The lower cutoff frequency is set by a high-value feedback resistance R_{FB} in parallel with feedback capacitor C_{FB}, $f_L = \frac{1}{2\pi R_{FB} C_{FB}}$. Achieving $f_L \approx 10mHz$ requires resistance on the order of teraohms. A MOS pseudo-resistor structure is used. In this design, the pseudo-resistor network provides approximately 3.76 TΩ of resistance.

The OTA core uses a thick oxide PMOS-input differential pair to reduce flicker noise. Device dimensions, multi-finger layout, and bias currents are tuned to balance transconductance, intrinsic gain, bandwidth, and power consumption. Increasing the number of fingers reduces thermal and flicker noise, but increases input capacitance, ultimately reducing bandwidth and gain. An output load capacitor C_L is used both to enhance stability and to set the upper cutoff frequency. Assuming dominant-pole behavior, the closed-loop cutoff is approximately $f_H = \frac{G_m}{2\pi A_v C_L}$, where G_m is the OTA transconductance and A_v is the midband closed-loop gain.

CMRR Measurement and DRL Operation

The CMRR of the ECG front end is obtained by separately simulating the differential-mode (DM) and common-mode (CM) gains at 60Hz using the full body and electrode models.

For the DM gain, the 60Hz interference current source is disabled, and the ECG differential source inside the body model is assigned an AC magnitude of 1V. The differential input voltage after the electrodes, $V_{in,dm} = V_{Right} - V_{Left}$, and the amplifier output $V_{out,dm}$ are measured in AC analysis to yield $A_{DM} = V_{out,dm}/V_{in,dm}$. For CM gain, the ECG source is set to AC = 0 while the 60Hz interference current source is enabled. First, with the DRL loop disconnected, the common-mode input voltage after the electrode is measured as $V_{in,CM} = \frac{V_{Right}+V_{Left}}{2}$. Second, the DRL loop is reconnected the resulting output amplitude $V_{out,CM}$ at 60Hz is measured. Following the course convention, the effective CM gain with DRL enabled is defined as

$A_{CM} = V_{out,cm}/V_{in,cm}$, and the CMRR at 60Hz is computed as $CMRR_{dB} = 20\log_{10}(A_{DM}/A_{CM})$, evaluated for both DRL-on and DRL-off cases.

Design Challenges and Simulation Methodology

Several challenges arose during design and verification. Increasing the OTA finger count generally reduced noise but degraded bandwidth and midband gain. Raising bias current improved linearity and bandwidth but risked violating the 4μW power constraint. Achieving stable DRL operation required careful tuning of its loop gain. Transistor lengths were increased to raise intrinsic gain, and the tail current source was adjusted to boost DRL loop gain while avoiding oscillation. Simulations were performed to verify common-mode suppression at 60Hz under the provided electrode model.

Properly driving the circuit for DM and CM simulations was critical because the DRL loop modifies the common-mode voltage. The transition from ideal components to PDK devices introduced parasitics that shifted cutoff frequencies and reduced gain until capacitor and pseudo-resistor values were readjusted. Monte Carlo simulations revealed sensitivity to mismatch. In Cadence, mismatch data was saved, and subsequent simulations reused the stored mismatch set so that DM and CM metrics were evaluated under similar device-mismatch conditions.

Conclusion

The final AFE meets the target differential gain, input impedance, bandwidth, noise, and CMRR specifications for low-power ECG acquisition. Key features include a 3.76 TΩ MOS pseudo-resistor, capacitive gain-setting architecture, realistic body and electrode modeling, and an optimized DRL loop for CMRR enhancement. Future work could incorporate chopper stabilization to further reduce noise, auto-zeroing for DC offset cancellation, improved pseudo-resistor linearity, and extension to a multistage architecture for larger output swing and robustness to mismatch.

Acknowledgments

The authors would like to thank Professor Mercier and Samuel Lahti for their guidance and responsiveness on Piazza.

References:

[1] H. Rezaee-Dehsorkh, N. Ravanshad, R. Lotfi, K. Mafinezhad, and A. M. Sodagar, "Analysis and design of tunable amplifiers for implantable neural recording applications," IEEE Journal on Emerging and Selected Topics in Circuits and Systems, vol. 1, no. 4, pp. 546–556, Dec. 2011

[2] P. "Mercier Lec8-9-10-BioAmp," ECE203, Fall Quarter 2025.

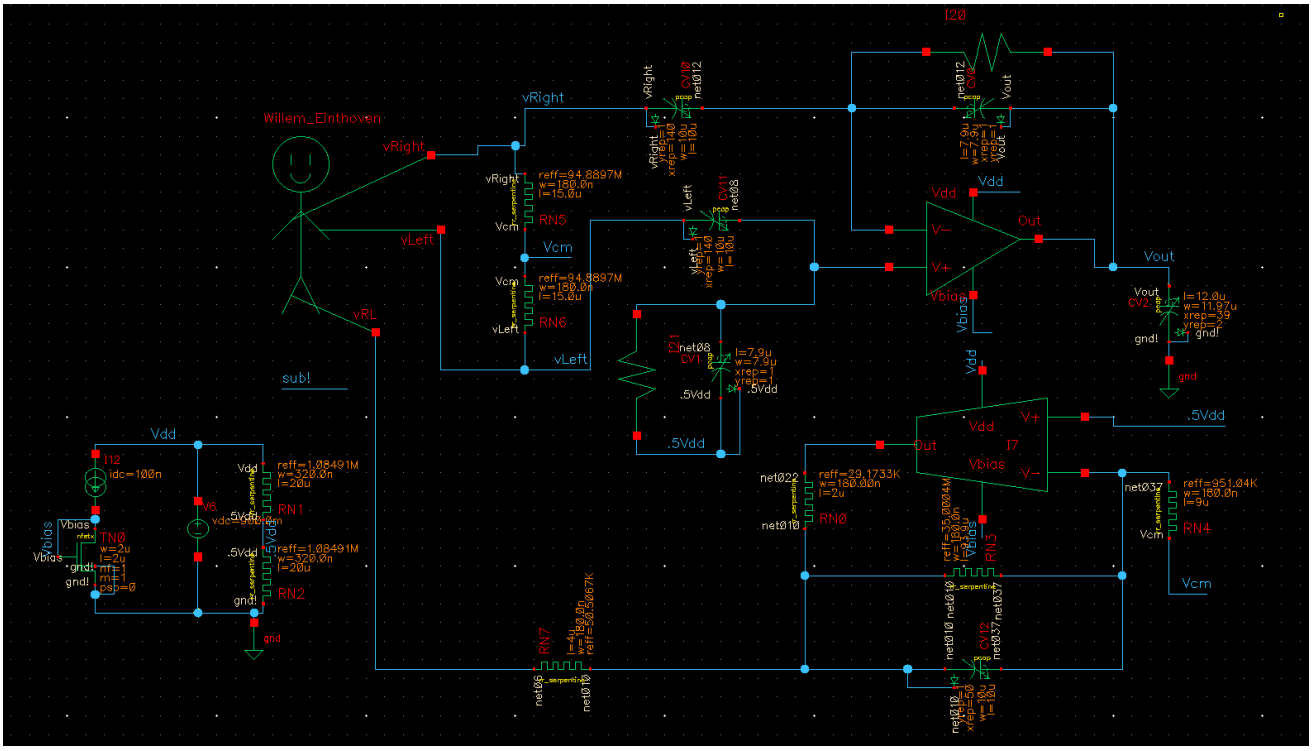

Figure 1.1 Full ECG Schematic

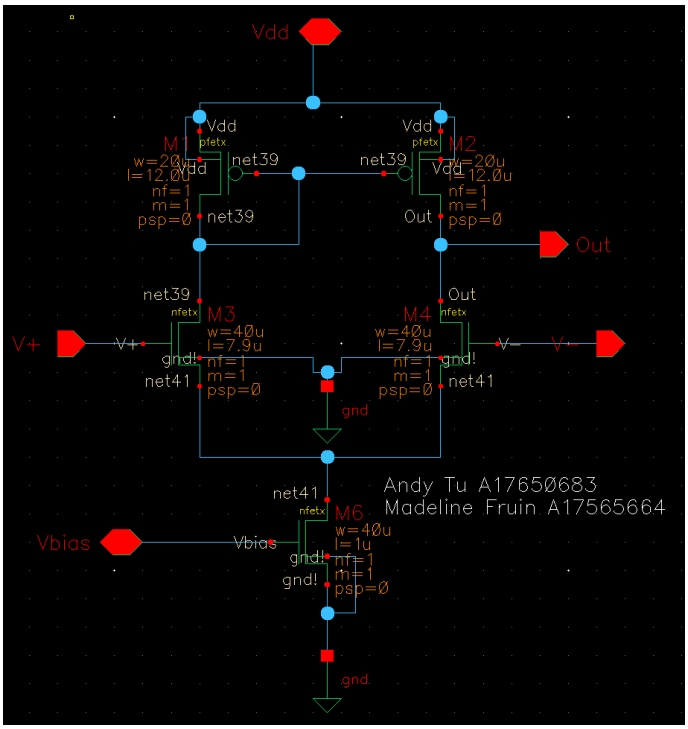

Figure 1.2 DRL Amplifier Schematic

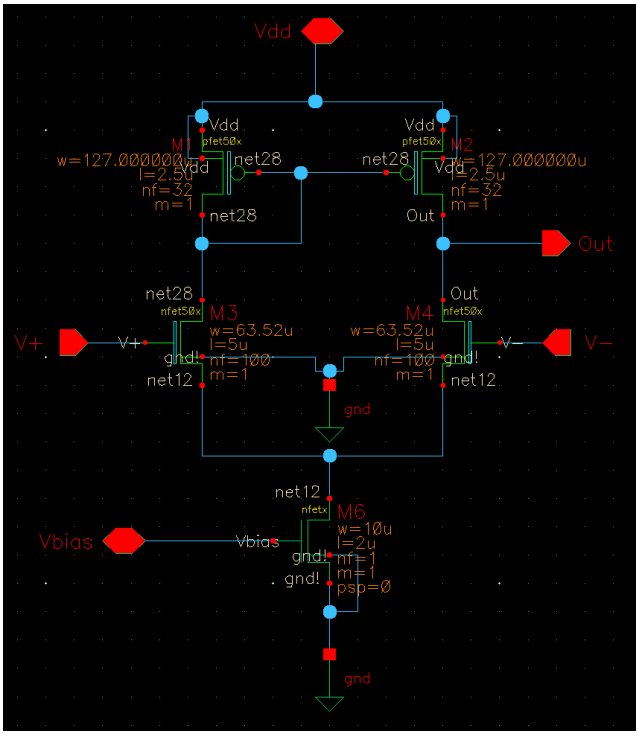

Figure 1.3 Single Stage Amplifier Schematic

Figure 1 | Circuit Schematics

CMRR with DRL

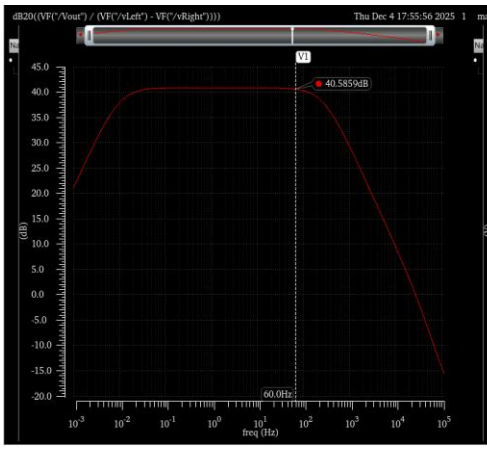

Figure 3.1: $A_{DM,dB}(60)$

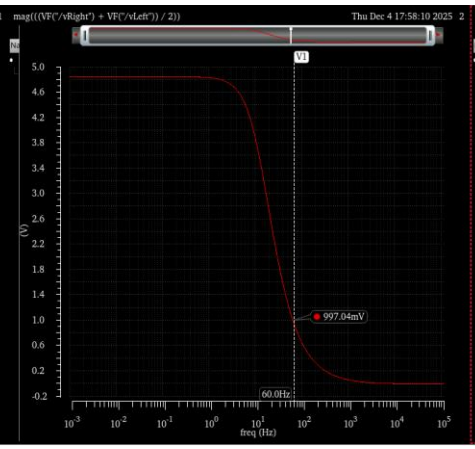

Figure 4.2: $V_{CM,in}(60)$

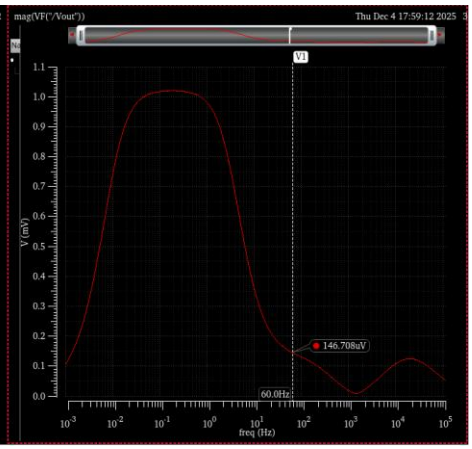

Figure 2.3: $V_{out,CM}(60)$

$$A_{DM,dB}(60)$$

$$A_{CM,dB} = \frac{|V_{out,cm}\ (60)|}{(|V_{cm,in}\ (60)|)}$$

$$Total\ CMRR_{dB}(60Hz) = A_{DM,dB}(60) - A_{CM,dB}\ (60)$$

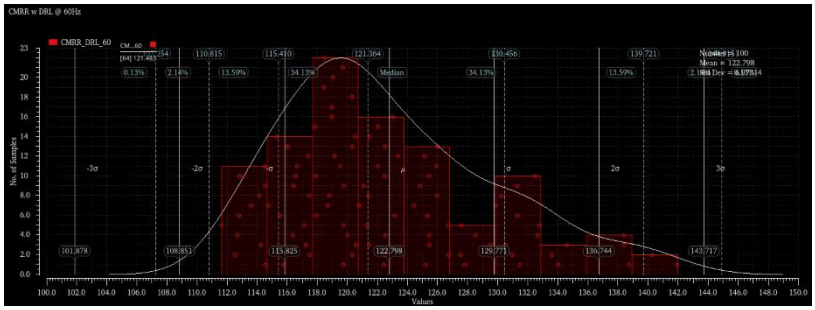

Figure 5.4: Monte Carlo Simulation Results: CMRR w/ DRL

Without DRL

S

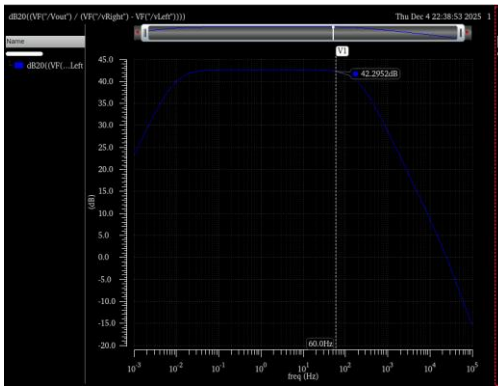

Figure 7.5: $A_{DM,in}(60)$

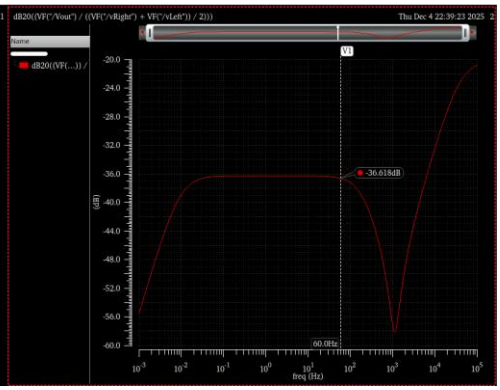

Figure 6.6: $A_{CM,in}(60)$

$$CMRR = 42.2952\ dB - (-36.618dB) = 78.9132\ dB$$

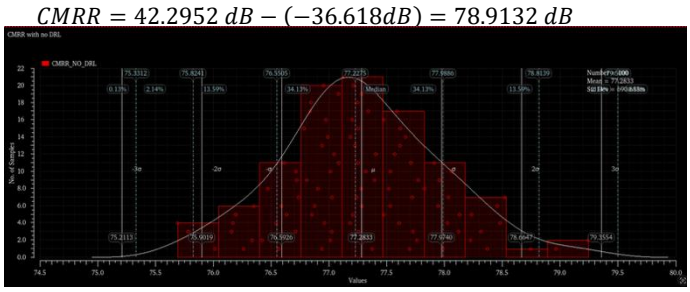

Figure 2.7: Monte Carlo Simulation Results: CMRR w/o DRL

Figure 2 | Common Mode Rejection Ratio (CMRR)

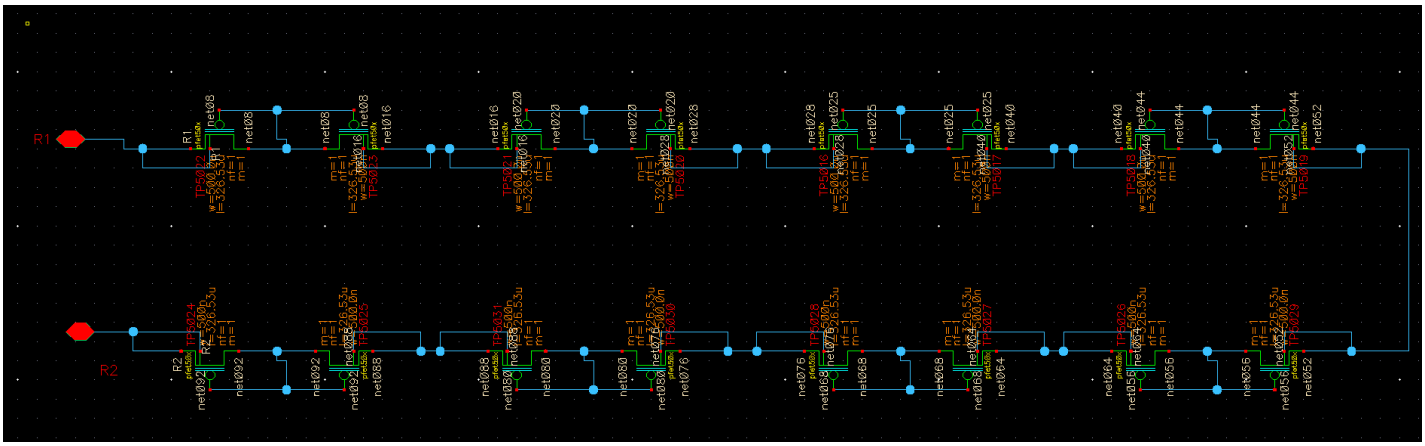

Figure 3.1 Pseudo Resistor Schematic

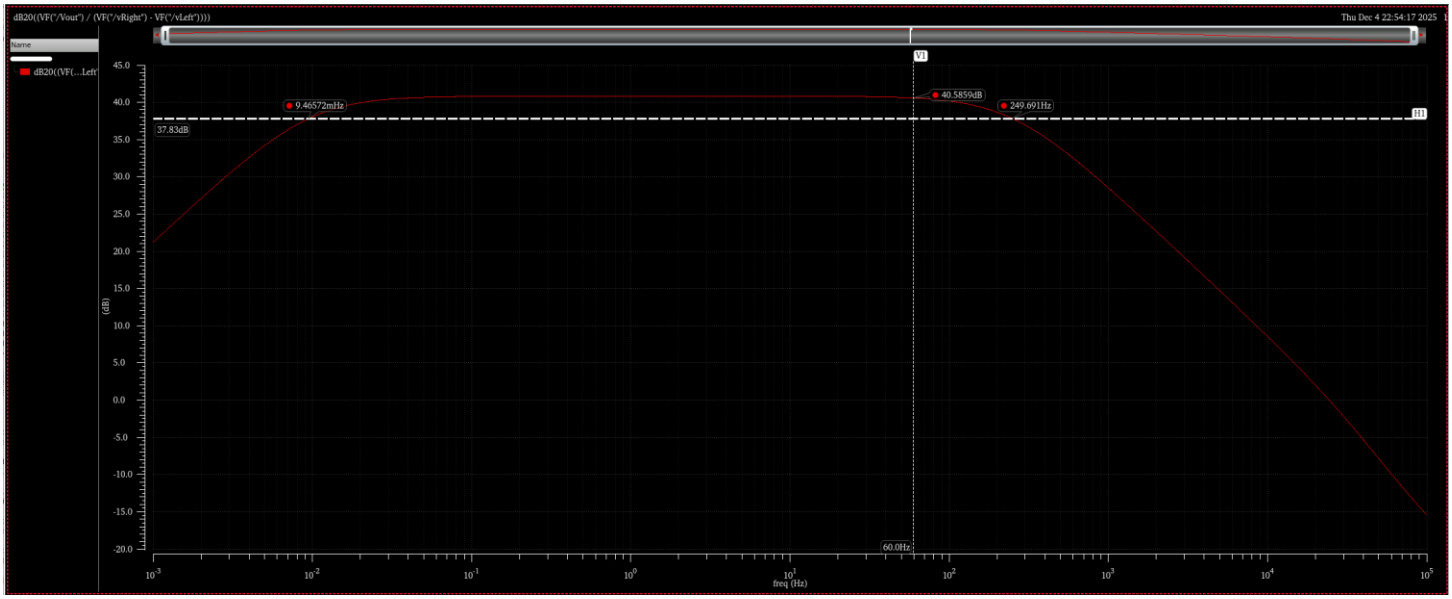

Figure 3.2 Bandwidth (9.46mHz to 249.691Hz)

Figure 3 | Pseudo Resistor Schematic and Bandwidth

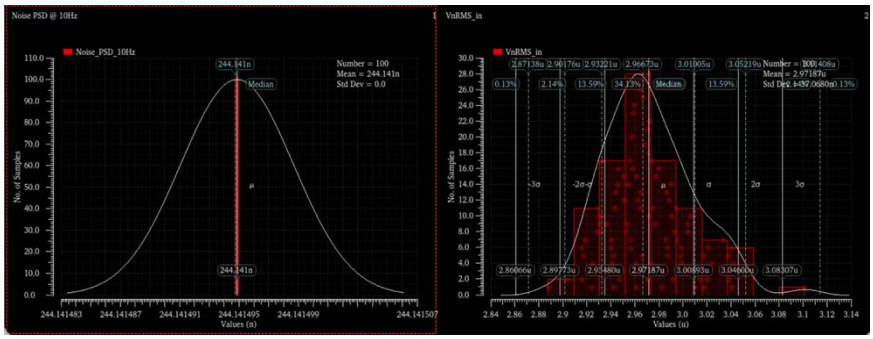

Figure 4.1 Noise PSD @ 10 Hz and Total Integrated input-referred noise

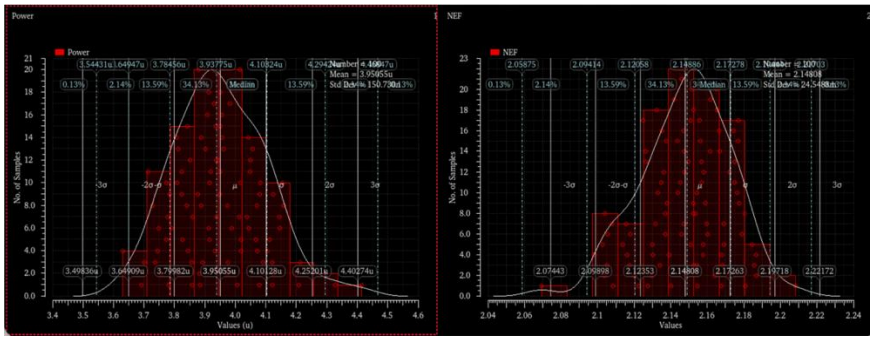

Figure 4.2 Power and NEF

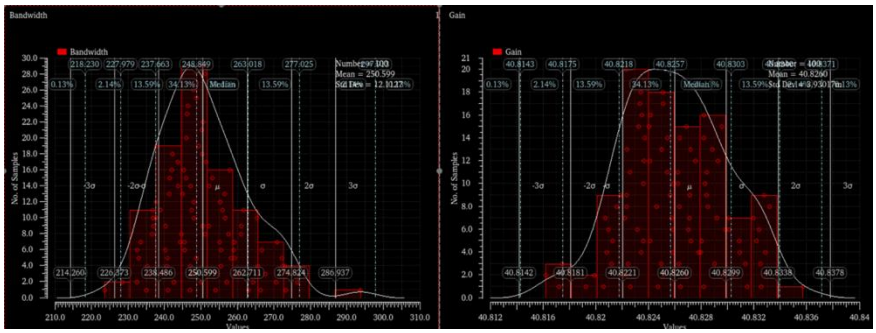

Figure 48.3 Bandwidth and Gain

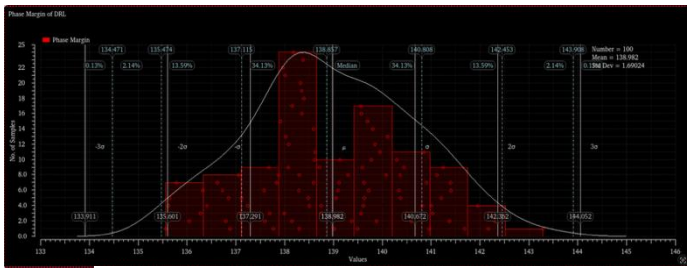

Figure 4.4 Phase Margin

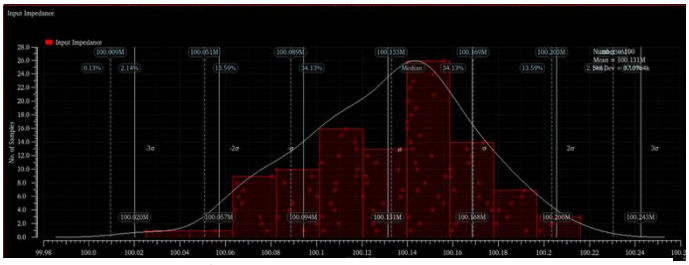

Figure 4.5 Input Impedance

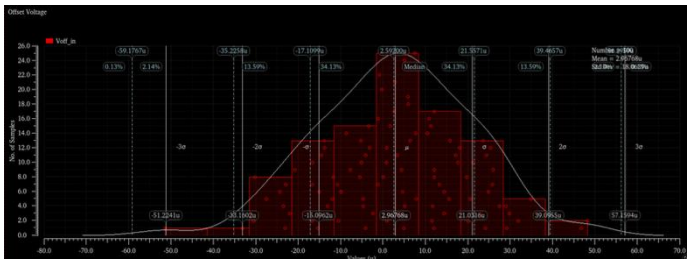

Figure 4.6 Input referred Offset Voltage

Figure 4 | Monte Carlo Simulation Results

	Ideal Performance (non-mismatched)	Worst-case mismatched performance	Best-case mismatched performance
Bandwidth	250.6 Hz	223.6 Hz	298.8 Hz
Gain	40.83 dB	40.82 dB	80.84 dB
Input-referred Noise PSD	$244.1\ nV/\sqrt{Hz}$	$244.1\ nV/\sqrt{Hz}$	$244.1\ nV/\sqrt{Hz}$
Total Integrated input-referred noise	$3.972\ \mu V$	$3.101\ \mu V$	$2.888\ \mu V$
Input impedance @ 10 Hz	$100.131\ M\Omega$	$100.243\ M\Omega$	$100.020\ M\Omega$
Power	$3.951\ \mu W$	$4.415\ \mu W$	$3.63\ \mu W$
CMRR w/o DRL @ 60Hz	79.3554 dB	75.2113 dB	79.3554 dB
CMRR w/ DRL @ 60 Hz	122.3 dB	101.878 dB	143.717 dB
DRL Phase Margin	138.9°	144.052°	133.911°
NEF	2.148	2.208	2.069
Total Amount of capacitance used	.967083298nF	–	–
Number of Voltage Sources used and their voltage	1 voltage Source used V = 900mV	–	–
Input referred Offset Voltage	$2.96\ \mu V$	$\pm 59.1767\ \mu V$	$0\ \mu V$

Figure 5 | Table

Device	Param	Noise Contribution	% Of Total
/Front_End_Amp/M4	Sfl	7.67479e-05	14.91
/Front_End_Amp/M3	Sfl	7.67408e-05	14.90
/Front_End_Amp/M3	Sth	6.61202e-05	11.06
/Front_End_Amp/M4	Sth	6.60543e-05	11.04
/Front_End_Amp/M2	Sth	6.33537e-05	10.16
/Front_End_Amp/M1	Sth	6.28842e-05	10.01
/Front_End_Amp/M2	Sfl	6.07898e-05	9.35
/Front_End_Amp/M1	Sfl	6.02785e-05	9.20
/TN0	fn	3.71778e-05	3.50
/Willem_Einthoven/R5	rn	3.10741e-05	2.44
/Willem_Einthoven/R7	rn	3.10713e-05	2.44
/Front_End_Amp/M6	fn	1.45441e-05	0.54
/Willem_Einthoven/R4	rn	8.63358e-06	0.19
/Willem_Einthoven/R6	rn	8.63248e-06	0.19
RN12.rma	rn	2.27523e-06	0.01
RN12.rmb	rn	2.27513e-06	0.01
RN11.rma	rn	2.25192e-06	0.01
RN11.rmb	rn	2.25189e-06	0.01
/Willem_Einthoven/R3	rn	1.82e-06	0.01
/Willem_Einthoven/R2	rn	9.10175e-07	0.00
/Willem_Einthoven/R1	rn	9.09828e-07	0.00
Front_End_Amp.M4.rsb	rn	7.26923e-07	0.00
Front_End_Amp.M3.rsb	rn	7.26899e-07	0.00
/TN0	id	6.64137e-07	0.00
Front_End_Amp.M2.rsb	rn	6.05856e-07	0.00

Integrated Noise Summary (in V) Sorted By Noise Contributors
Total Summarized Noise = 0.000198785
Total Input Referred Noise = 2.93183e-06
The above noise summary info is for noise data

Figure 6.1 Noise summary with electrodes

Device	Param	Noise Contribution	% Of Total
/Front_End_Amp/M4	Sfl	7.67479e-05	14.91
/Front_End_Amp/M3	Sfl	7.67408e-05	14.90
/Front_End_Amp/M3	Sth	6.61202e-05	11.06
/Front_End_Amp/M4	Sth	6.60543e-05	11.04
/Front_End_Amp/M2	Sth	6.33537e-05	10.16
/Front_End_Amp/M1	Sth	6.28842e-05	10.01
/Front_End_Amp/M2	Sfl	6.07898e-05	9.35
/Front_End_Amp/M1	Sfl	6.02785e-05	9.20
/TN0	fn	3.71778e-05	3.50
/Front_End_Amp/M6	fn	1.45441e-05	0.54
RN12.rma	rn	2.27523e-06	0.01
RN12.rmb	rn	2.27513e-06	0.01
RN11.rma	rn	2.25192e-06	0.01
RN11.rmb	rn	2.25189e-06	0.01
Front_End_Amp.M4.rsb	rn	7.26923e-07	0.00
Front_End_Amp.M3.rsb	rn	7.26899e-07	0.00
/TN0	id	6.64137e-07	0.00
Front_End_Amp.M2.rsb	rn	6.05856e-07	0.00
Front_End_Amp.M1.rsb	rn	6.00705e-07	0.00
CV11.rtop	rn	4.31337e-07	0.00
CV10.rtop	rn	4.31273e-07	0.00
Front_End_Amp.M2.rgate	rn	4.06879e-07	0.00
Front_End_Amp.M1.rgate	rn	4.03424e-07	0.00
RD2.rea1	fn	3.86288e-07	0.00
RD1.rea1	fn	3.86288e-07	0.00

Integrated Noise Summary (in V) Sorted By Noise Contributors
Total Summarized Noise = 0.000198785
Total Input Referred Noise = 2.93183e-06
The above noise summary info is for noise data

Figure 6.2 Noise Summary without electrodes

Figure 6 | Noise Summary